

Team: Change Detectors

Abstract

This project focuses on designing and developing a role-based Insurance Policy Management System using Angular as the frontend framework and JSON Server as a simulated backend. The main objective of the project is to create a web application that manages insurance policies, customer onboarding, document uploads, and claims processing within a single platform.

In this system, an Admin user is responsible for creating and managing insurance policies and processing customer claims. Customers can register, log in, view available policies, purchase insurance policies, upload required KYC documents, file insurance claims, and track the status of their claims. Agents are included to assist in managing assigned customers and tracking policy sales and commissions. All application data is stored in a local db.json file, and REST API communication is implemented using JSON Server. Document handling is simulated by storing uploaded files in a local uploads folder and saving file references in the database.

The project is developed following a modular Angular architecture with separate components and services for each functional module. This implementation provides practical exposure to frontend development, REST API integration, role-based authentication, and real-world insurance workflow simulation.

Technology Stack

Frontend: Angular (Framework) ,TypeScript (Language) , HTML5 , CSS3 , Tailwind CSS

Backend Simulation: JSON Server (REST API simulation)

Authentication: JWT-based login simulation , Role-based route protection

File Storage: Local uploads/ directory for document storage

Version Control: Git & GitHub

Members' Roles and Responsibilities

1. Member 1 (Srujana) – Customer Management Module

Responsibilities:

- Implementation of customer registration and profile management.
- Display of customer personal information and contact details.
- Implementation of KYC document upload and status display.
- Development of customer policy portfolio page.
- Display of renewal schedule and communication preferences.
- Integration of customer data with JSON Server.
- Design and development of the Customer Module.

Design and development of the Customer Module.

2. Member 2 (Himapriya) – Policies Management Module

Responsibilities:

- Design and development of the Policy Management Module.
- Implementation of policy CRUD operations (Add, View, Edit, Delete).
- Display of available insurance policies to customers.
- Implementation of policy browsing and filtering features.
- Integration of policy data with JSON Server.
- Linking purchased policies to customer profiles.

3. Member 3 (Bharath) – Authentication and Claims Processing Module

Responsibilities:

- Implementation of Authentication Module (Login and Registration).
- Role-based redirection and route protection.
- JWT token simulation and session handling.
- Development of Claim Filing functionality for customers.
- Implementation of claim tracking interface.
- Development of Admin claim approval and rejection features.
- Integration of claims data with JSON Server.

4. Member 4 (Anand) – Dashboard Module

Responsibilities:

- Design and development of role-based dashboards (Admin, Agent, Customer).
- Implementation of dashboard layouts and navigation.
- Display of key statistics and summary information.
- UI design and responsive layout implementation.
- Integration of all modules into a unified dashboard experience.

Core Application Modules

1. Authentication Module

The Authentication Module manages user access to the system. It allows users to register and log in based on their roles.

Key Functionalities:

User Registration (Customer and Agent)

User Login

Role-based redirection (Admin, Agent, Customer dashboards)

JWT token simulation for session handling

Route guards to protect authorized pages

2. Admin Module

The Admin Module provides complete control over system operations and data management.

Key Functionalities:

Admin Dashboard overview

Policy Management (Add, View, Edit, Delete policies)

Claims Management (View, Approve, Reject claims)

Customer and Agent monitoring

3. Customer Module

The Customer Module enables users to interact with insurance services.

Key Functionalities:

Customer Registration and Profile Management

Browse and purchase insurance policies

Upload KYC documents

File insurance claims

Track claim status

View policy portfolio and renewal details

4. Agent Module

The Agent Module supports intermediaries who assist customers.

Key Functionalities:

View assigned customers

Track policies sold

View commission details

Maintain customer communication logs

5. Policy Management Module

This module handles insurance policy data and operations.

Key Functionalities:

Display available policies

Filter and search policies

Policy CRUD operations (Admin)

Policy purchase process (Customer)

6. Claims Processing Module

The Claims Module manages the end-to-end claim process.

Key Functionalities:

Claim registration by customers

Document upload for claims

Claim status tracking

Claim approval or rejection by Admin

Remarks and communication updates

7. Document Management Module

This module manages customer and claim-related documents.

Key Functionalities:

Upload KYC and claim documents

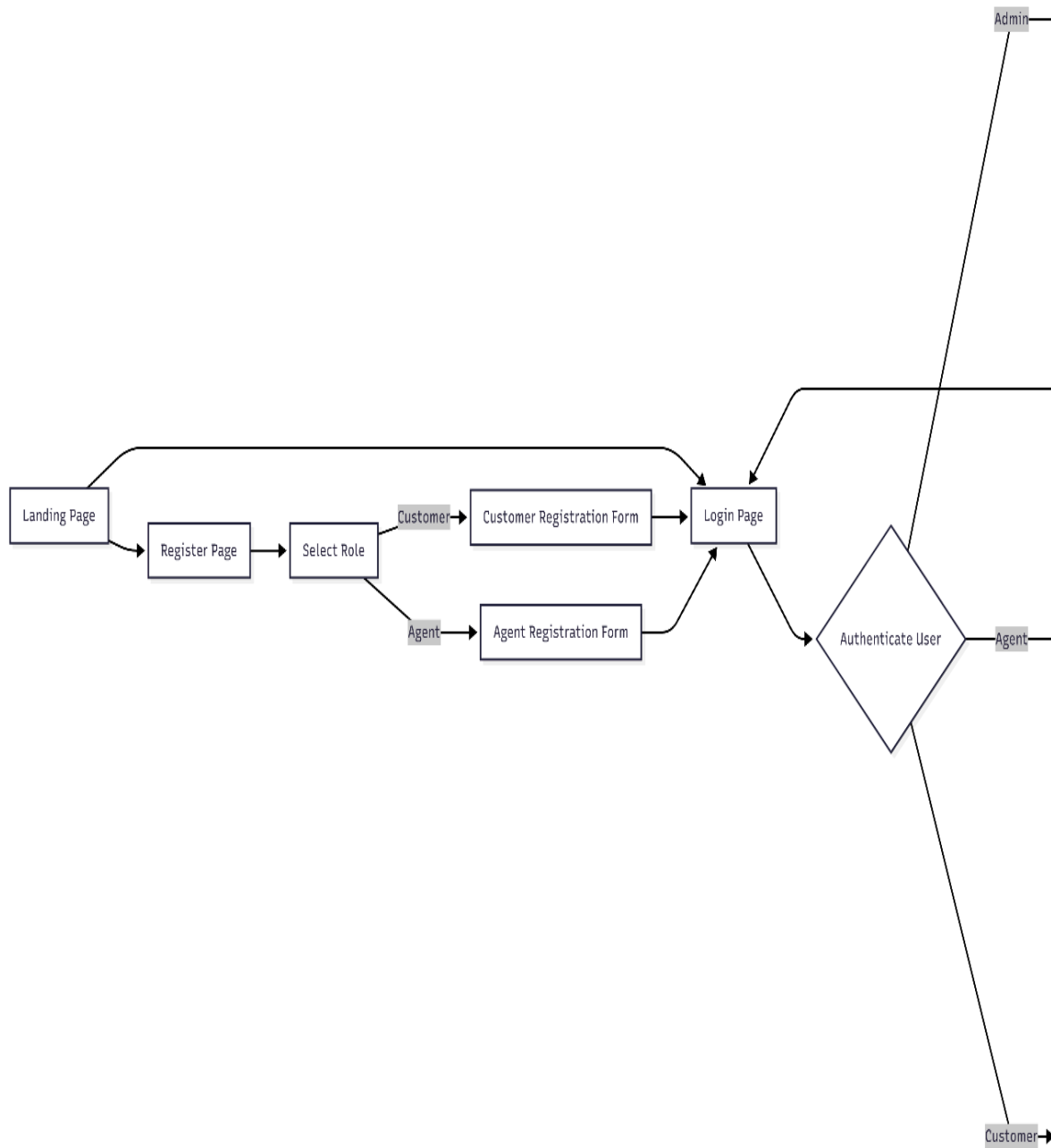
Document categorization

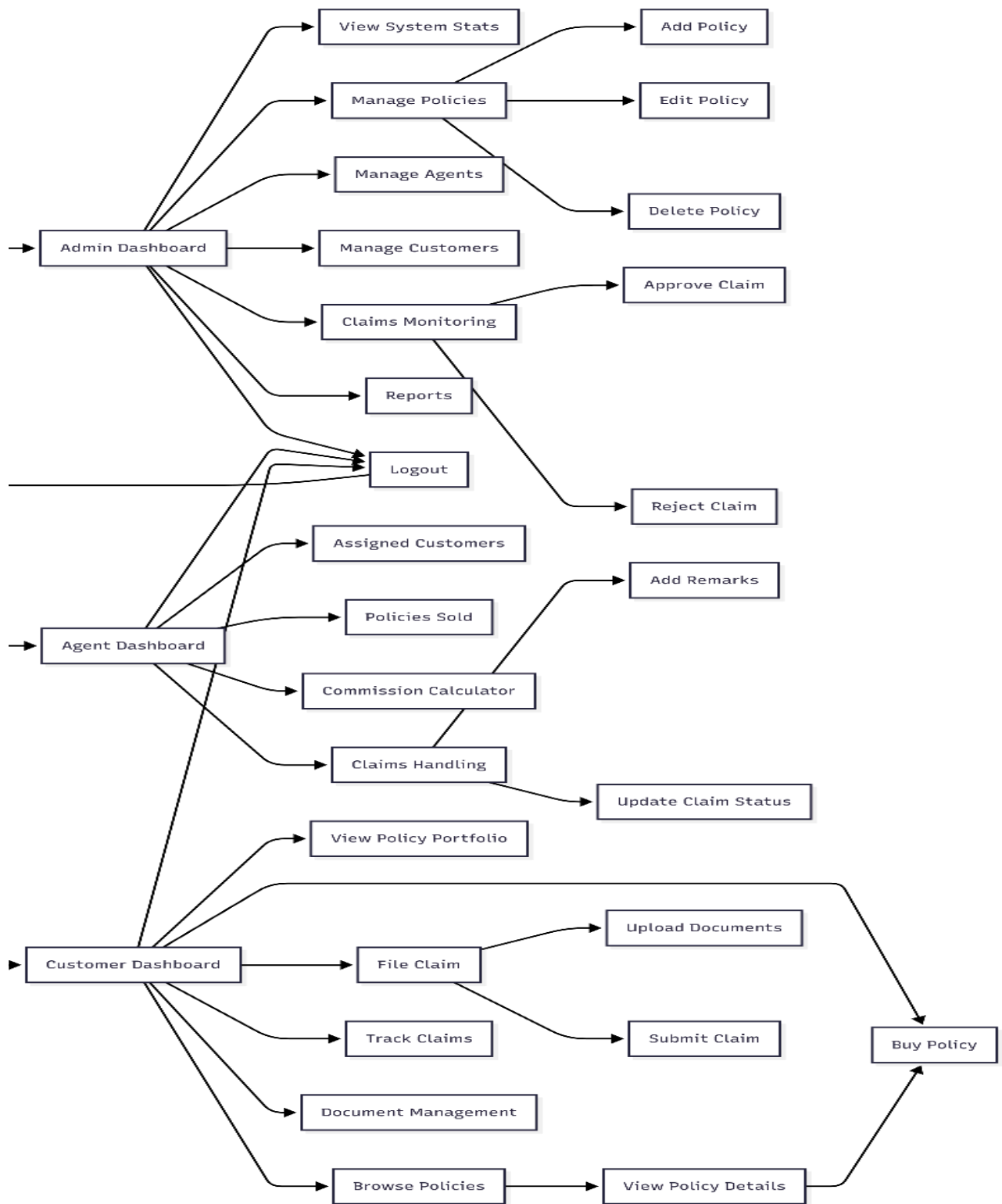
Store file references in database

Display uploaded documents

Expiry status indication

System Architecture





Database Schema Design (JSON Server)

1. Users Collection

Stores login and role information for all system users.

```
{ "users" : [  
  {  
    "id": 1,  
    "username": "admin",  
    "email": "admin@insurance.com",  
    "password": "admin123",  
    "role": "admin"  
  },  
  {  
    "id": 2,  
    "username": "agent1",  
    "email": "agent1@insurance.com",  
    "password": "agent123",  
    "role": "agent"  
  },  
  {  
    "id": 3,  
    "username": "customer1",  
    "email": "customer1@insurance.com",  
    "password": "cust123",  
    "role": "customer"  
  }  
]  
}
```

2. Policies Collection

Stores insurance policy details created by Admin.

```
{
  "policies": [
    {
      "id": "POL-001",
      "name": "Health Insurance Plan",
      "type": "Health",
      "premium": 12000,
      "coverage": 500000,
      "durationDays": 365,
      "description": "Comprehensive health coverage"
    }
  ]
}
```

3. Customers Collection

Stores customer profile and purchased policy details.

```
{
  "customers": [
    {
      "id": "CUST-001",
      "userId": 3,
      "fullName": "John Doe",
      "phone": "9876543210",
      "address": "Hyderabad",
      "kycStatus": "Verified",
      "communicationPreference": "Email",
      "policyIds": ["POL-001"],
      "policyStartDate": "2026-01-10"
    }
  ]
}
```


4. Agents Collection

Stores agent information and assigned customers.

```
{
  "agents": [
    {
      "id": "AGT-001",
      "userId": 2,
      "assignedCustomers": ["CUST-001"],
      "commissionRate": 10
    }
  ]
}
```

5. Claims Collection

Stores insurance claim records submitted by customers.

```
{
  "claims": [
    {
      "id": "CLM-001",
      "policyId": "POL-001",
      "customerId": "CUST-001",
      "claimType": "Health",
      "amount": 50000,
      "status": "Pending",
      "documents": ["bill.pdf", "report.jpg"],
      "remark": ""
    }
  ]
}
```

6. Documents Collection

Stores uploaded document references.

```
{
  "documents": [
    {
      "id": "DOC-001",
      "customerId": "CUST-001",
      "documentType": "Aadhar",
      "fileName": "aadhar.pdf",
      "expiryDate": "2030-05-10"
    },
    {
      "id": "DOC-002",
      "customerId": "CUST-001",
      "documentType": "Medical Bill",
      "fileName": "bill.pdf"
    }
  ]
}
```

7. Communication Logs Collection

Stores notes added by agents for customer interactions.

```
{
  "communications": [
    {
      "id": "COM-001",
      "agentId": "AGT-001",
      "customerId": "CUST-001",
      "date": "2026-02-15",
      "message": "Called customer regarding claim document update."
    }
  ]
}
```